

Puducherry, India | jophitakristen523@wec.edu.in | linkedin.com/in/jophita-kristen

Information Science & Engineering undergraduate focusing on Explainable AI, Computer Vision, Machine Learning systems, and Full-Stack Engineering. Proven track record in hackathons, competitive academics, and prototype development.

EDUCATION

|                                                                                                                                                |                                 |
|------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------|
| Women's Engineering College, Lawspet, Puducherry                                                                                               | 2023 - 2027 (Expected)          |
| B.Tech in Information Science & Engineering                                                                                                    | CGPA: 8.78 / 10.0 (Rank 1 / 15) |
| Relevant Coursework: Data Structures & Algorithms, Object-Oriented Programming, Database Management Systems, Machine Learning, Computer Vision |                                 |
| St. Joseph of Cluny Higher Secondary School, Lawspet                                                                                           | 2021 - 2023                     |
| Higher Secondary Certificate (HSC - 12th Standard)                                                                                             | Percentage: 90.67%              |
| St. Joseph of Cluny Higher Secondary School, Lawspet                                                                                           | 2020 - 2021                     |
| Secondary School Leaving Certificate (SSLC - 10th Standard)                                                                                    | Percentage: 98.60%              |

TECHNICAL SKILLS

|                                                                                                                 |
|-----------------------------------------------------------------------------------------------------------------|
| Programming Languages: C, C++, Python, Java, JavaScript, TypeScript, SQL, HTML5, CSS3                           |
| Frameworks & Libraries: React, Node.js, Express, Flask, Tailwind CSS, Scikit-learn, OpenCV                      |
| Databases & Storage: MongoDB, MySQL, PostgreSQL                                                                 |
| Tools & Environments: Git, GitHub, VS Code, MATLAB, Simulink, Google Earth Engine, Figma                        |
| Domains & Competencies: Explainable AI, Computer Vision, Machine Learning, Full-Stack Development, RESTful APIs |

SELECTED ENGINEERING PROJECTS

|                                                                                                                                                                                                                        |                           |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------|
| Explainable AI for Diabetic Retinopathy Screening                                                                                                                                                                      | SIH 2026 • IN DEVELOPMENT |
| Tech: MATLAB, Simulink, ResNet-50, Grad-CAM                                                                                                                                                                            |                           |
| Developing an explainable clinical triage system for retinal fundus imaging. Integrates Grad-CAM activation mapping to highlight microaneurysms and hemorrhages, giving clinicians interpretable diagnostic reasoning. |                           |
| Smart Hospital AI Suite (CareFlow + MediCareConnect)                                                                                                                                                                   | Healthcare System         |
| Tech: Node.js, Express, MongoDB, REST APIs, JWT                                                                                                                                                                        |                           |
| Engineered modular healthcare backend services providing high-availability RESTful APIs for patient intake, medical record management, and doctor appointment scheduling with document persistence.                    |                           |
| ResuMatch Ultra — Candidate Semantic Scoring Engine                                                                                                                                                                    | HR Analytics              |
| Tech: Python, Scikit-learn, TF-IDF, Cosine Similarity                                                                                                                                                                  |                           |
| Built an automated resume evaluation engine that parses candidate documents, generates TF-IDF feature vectors, and computes cosine similarity against job descriptions to benchmark skill alignment.                   |                           |
| EncroWatch — Satellite Land Boundary Analysis                                                                                                                                                                          | Geospatial AI             |
| Tech: Python, Flask, Google Earth Engine, GIS Datasets                                                                                                                                                                 |                           |
| Created a satellite-based land monitoring tool combining Google Earth Engine and Flask to detect temporal vegetation and infrastructure encroachments across sensitive municipal boundaries.                           |                           |
| Scanline — Browser-Based Smart Checkout & Multi-Item Detection                                                                                                                                                         | In Testing                |
| Tech: Camera Capture API, Object Detection Model, Product Catalog DB                                                                                                                                                   |                           |
| Browser-based checkout prototype that detects multiple items laid on a flat surface using computer vision and calculates totals without barcode scanners or manual entry.                                              |                           |

HONORS & COMPETITIVE ACHIEVEMENTS

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|------------------------------------------------------------------------------------------|------|
| [1st Place] IDEATHON 2024 — Cash Prize: Rs. 10,000 — Women's Engineering College         | 2024 |
| [2nd Place] Brain Wave 2024 — Cash Prize: Rs. 3,000 — Inter-College Innovation Challenge | 2024 |
| [Performer] Code Fiesta 2024 — Algorithm Speed Run — Technical Symposium                 | 2024 |
| [Scholar] Academic Excellence Award — B.Tech Year 1 Rank 1 — Academic Directorate        | 2024 |

LEADERSHIP & CAMPUS INVOLVEMENT

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|--------------------------------------------------------------|----------------|
| Student Coordinator — Department Technical Club, WEC         | 2024 - Present |
| Workshop Lead — Peer Coding & Problem Solving Cohort         | 2024           |
| Volunteer — National Service Scheme (NSS) Community Outreach | 2023 - Present |

CERTIFICATIONS & PROFESSIONAL WORKSHOPS

|                                               |      |
|-----------------------------------------------|------|
| Machine Learning with Python — Coursera / IBM | 2024 |
| Full Stack Web Development Bootcamp — Udemy   | 2024 |
| Computer Vision Fundamentals — Great Learning | 2024 |

